23-200-0105A
Basic Mechanical Engineering

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Syllabus

Module I

Introduction: Role of Mechanical Engineering in Industries and Society- Emerging Trends and Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors.

Introduction to thermodynamic laws, power generating devices: Boilers, Turbines (Steam & Gas), IC engines: Components and Working Principles of two stroke petrol engine and 4-Strokes Petrol and Diesel Engines, Application of IC Engines. (Elementary ideas only no numerical problems).

Introduction to power consuming devices: Refrigerator, types and properties of refrigerants, working of domestic refrigerators, Air-conditioning systems, Windows and Split systems (only elementary ideas, no numerical problems)

Module II

Introduction to power transmission systems: Belts, chain, and Gear drives, types and application, (numerical problems related to simple power calculations only).

Energy: Introduction and applications of Energy sources like Fossil fuels, nuclear fuels, Hydel, Solar, wind, and bio-fuels, Environmental issues like Global warming and Ozone depletion.

Modern fuel injection systems in CI and SI engines: CRDi, MPFI systems, cooling and lubricating systems in two stroke and four stroke engines. (Only elementary ideas with block diagrams).

Insight into Future Mobility: Electric and Hybrid Vehicles, Components of Electric and Hybrid Vehicles. Advantages and disadvantages of EVs and Hybrid vehicles.

Module III

Introduction to engineering materials: composite and smart materials.

Joining Processes: Soldering, Brazing and Welding, Definitions, classification of welding process, Arc welding, Gas welding and types of flames.

Machine Tool Operations: Working Principle of lathe, Lathe operations: Turning, facing, knurling. Working principles of Drilling Machine, drilling operations: drilling, boring, reaming. Working of Milling Machine, Milling operations: plane milling and slot milling.

(No sketches of machine tools, sketches to be used only for explaining the operations).

Introduction to Advanced Manufacturing Systems: Introduction, components of CNC, advantages and applications of CNC, 3D printing.

Syllabus…

Module IV
Introduction to Mechatronics and Robotics: open-loop and closed-loop mechatronic systems. Classification based on robotics configuration: polar cylindrical, Cartesian coordinate and spherical. Application, Advantages and disadvantages.
Automation in industry: Definition, types – Fixed, programmable and flexible automation, basic elements with block diagrams, advantages.
Introduction to IOT: Definition and Characteristics, Physical design, protocols, Logical design of IoT, Functional blocks, and communication models.

References:

1. Jonathan Wickert and Kemper Lewis. An Introduction to Mechanical Engineering, Third Edition, Cengage Learning (2012).
2. Hazra Choudhry and Nirzar Roy. Elements of Workshop Technology (Vol. 1 and 2), Media Promoters and Publishers Pvt. Ltd. (2010).
3. V. Ganesan. Internal Combustion Engines, 4th edition, Tata McGraw Hill Education (2017).
4. Appu Kuttan K K. Robotics Volume 1, I K. International Publishing House Pvt Ltd (2013).
5. SRN Reddy, Rachit Thukral and Manasi Mishra. Introduction to Internet of Things: A Practical Approach, ETI Labs (2021).

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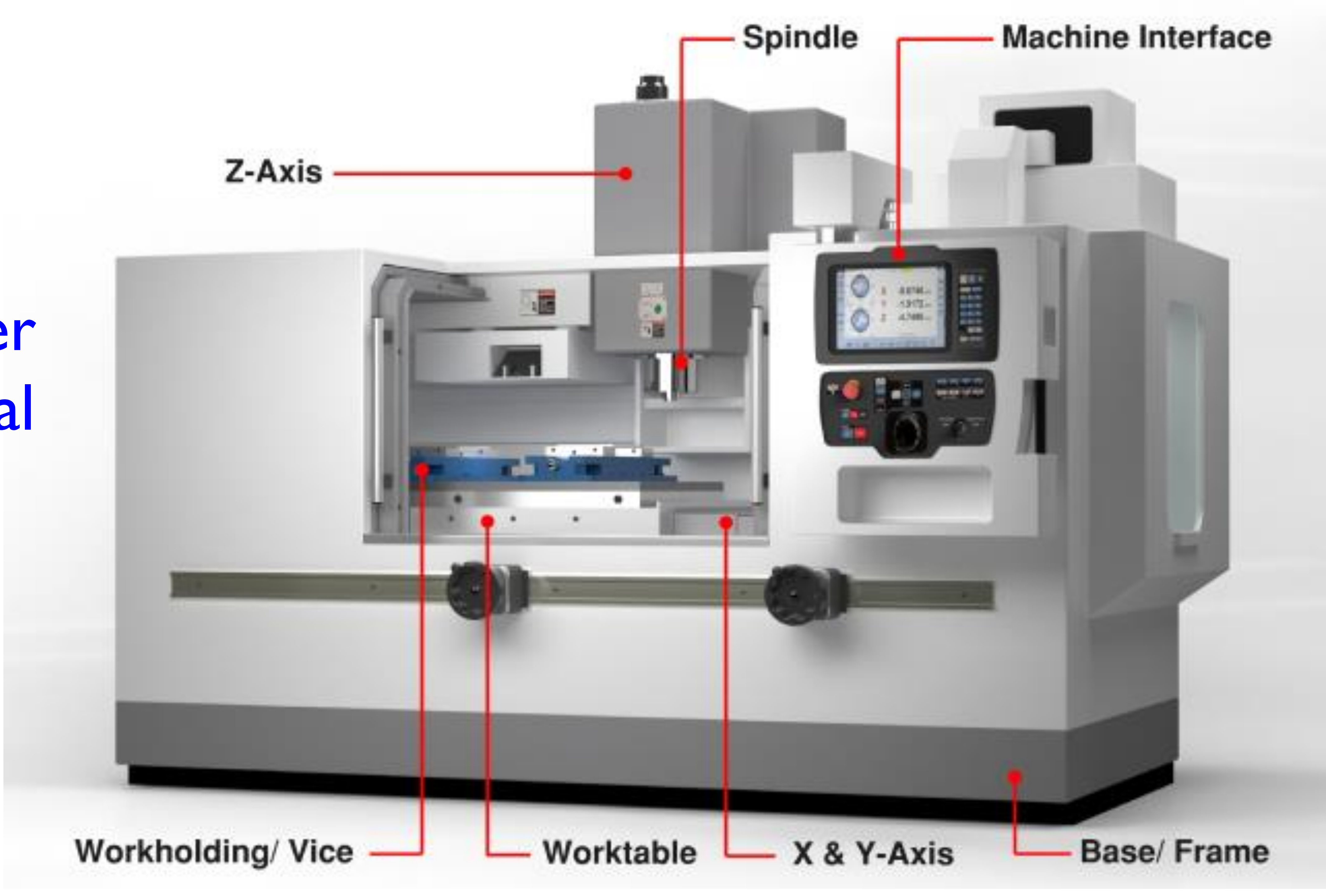
Computer Numerical Control



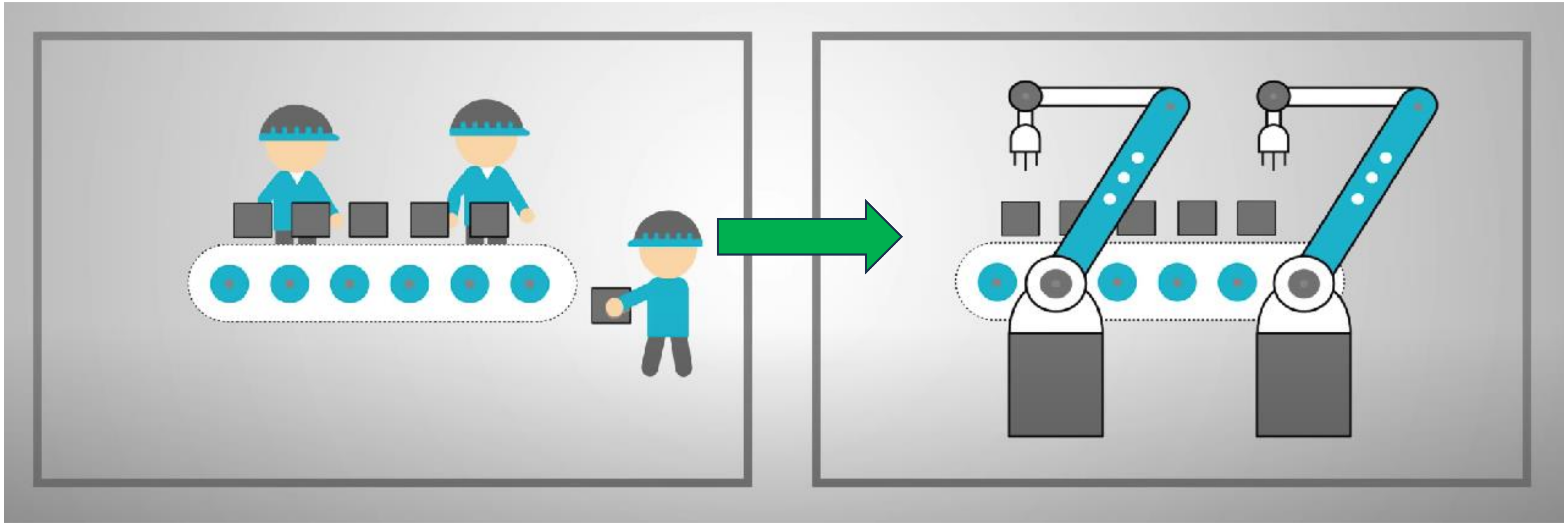
Computer Numerical Control

- Numerical Control:- a programmable automation in which process is controlled by Numbers, Letters, and symbols
- CNC Machining is a process used in the manufacturing sector that involves the use of computers to control machine tools like lathes, mills and grinders.

Why CNC?

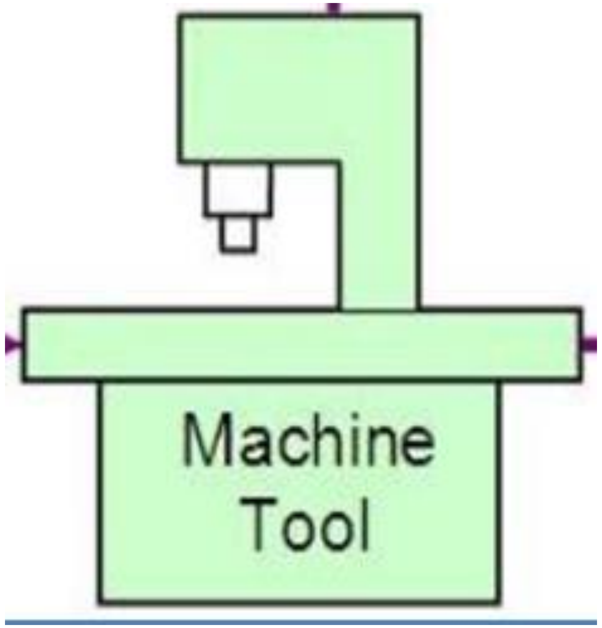
- To manufacture complex curved geometries in 2D or 3D was extremely expensive by mechanical means
- Machining components with high Repeatability and Precision
- Unmanned machining operations

Automation?



Automation can be defined as the technology by which a process or procedure is accomplished without human assistance.

Example: Drilling operation



Parts with better accuracy

Control Systems

Open Loop

- No access to real time data
- No immediate corrective action taken incase of system disturbance
- Ex: Wirecut machines

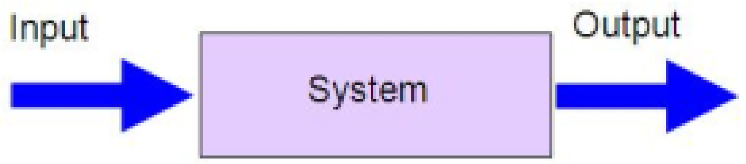


Fig.1-2(a) Block Diagram of an Open Loop System

Closed loop

- Feedback device closely monitor output
- Corrective action is taken if there is some disturbances
- Ex: most of the CNCs now a days are closed loop system

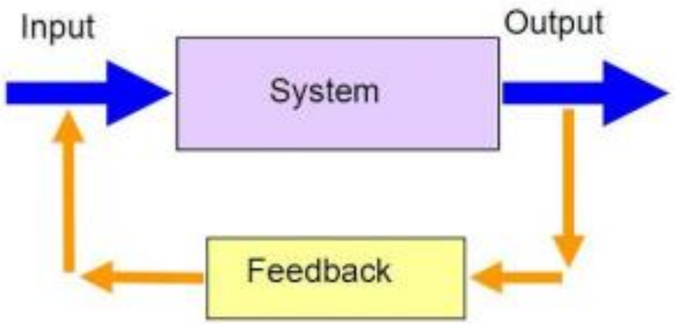
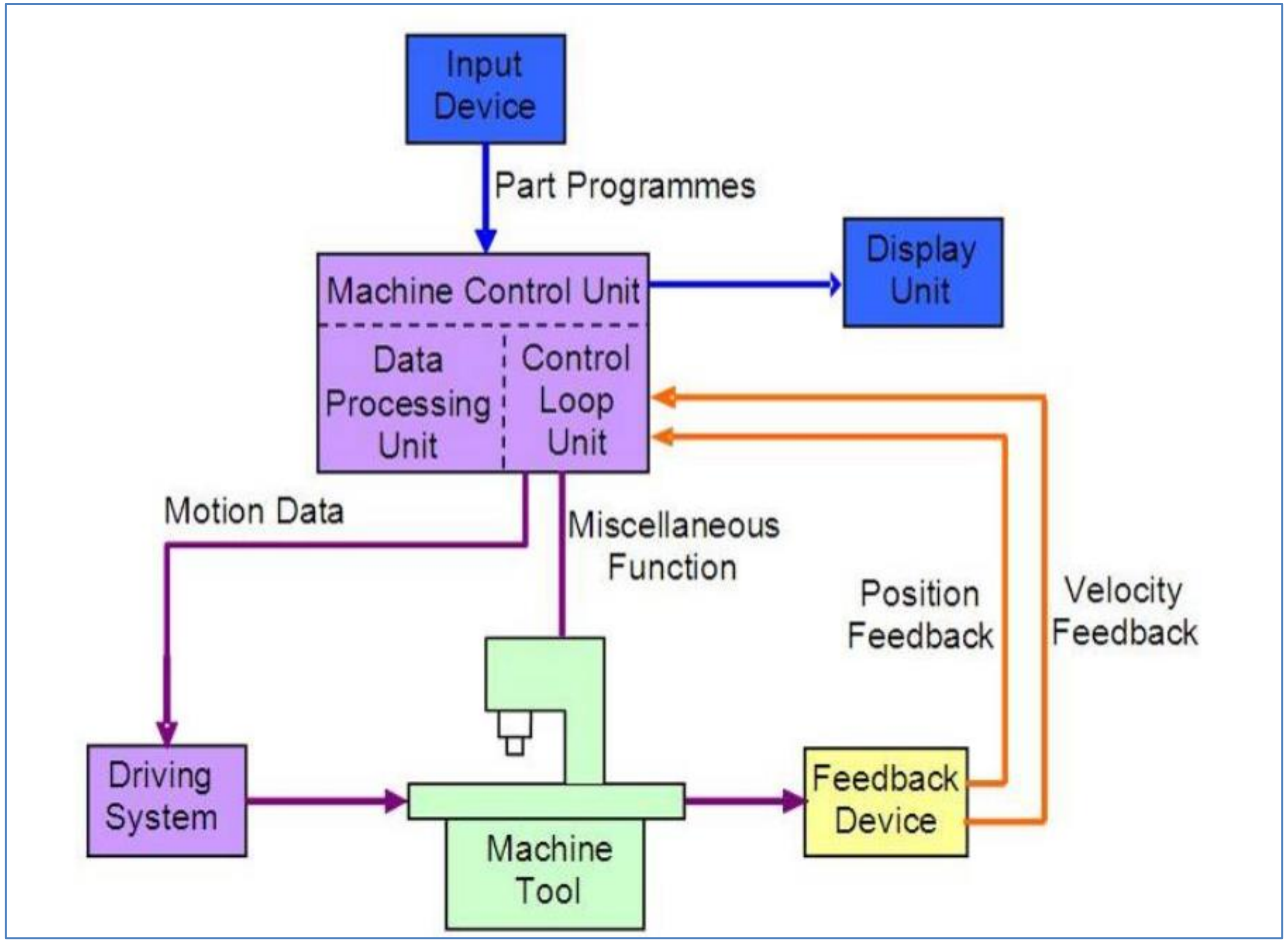


Fig.1-2(b) Block Diagram of a Close Loop System

Elements of CNC

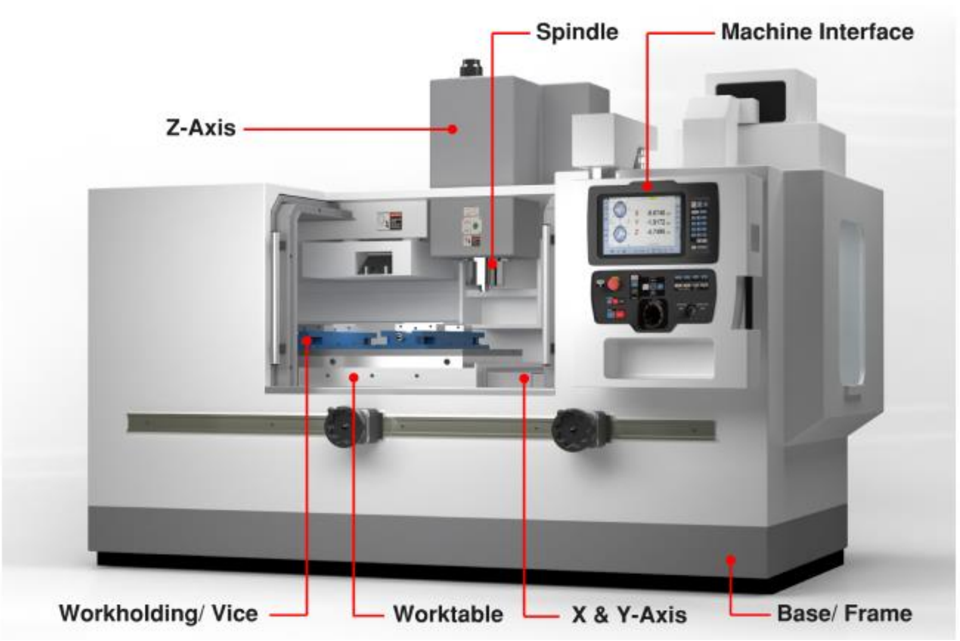


Elements of CNC

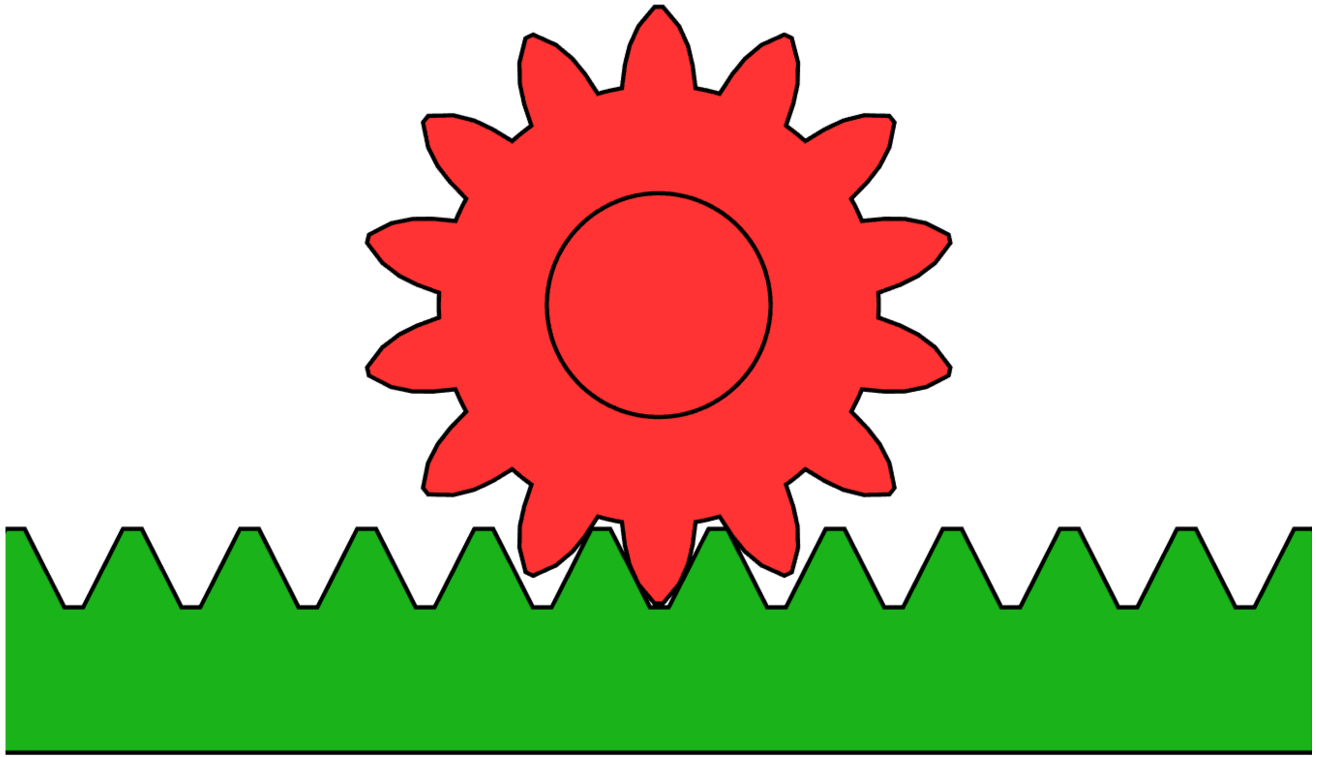
- **Input Device**:- Input to the CNC machine is given through this element, typically USBs, CDs or through inbuilt computer
- **Machine Control Units** (MCUs):- Heart of CNC system
  - Data Processing Unit (DPU):- Part program converted to internal machine codes
  - Control Loop Unit (CLU):- Data from DPU converted to electrical signals to control driving system.
- **Machine Tool**:- Material removal operation is carried out here
- **Driving system**:- Provide controlled motion to CNC elements, Electric/hydraulic motors for driving the machine

Elements of CNC

- **Feedback device**:- Updating positional values and speed of the axes need to be contently updated
  - Positional Feedback
  - Velocity Feedback
- **Display Units**:- Interactive unit between machine and operator



Rotary motion to linear motion



Rack and Pinion arrangement

Applications of CNC

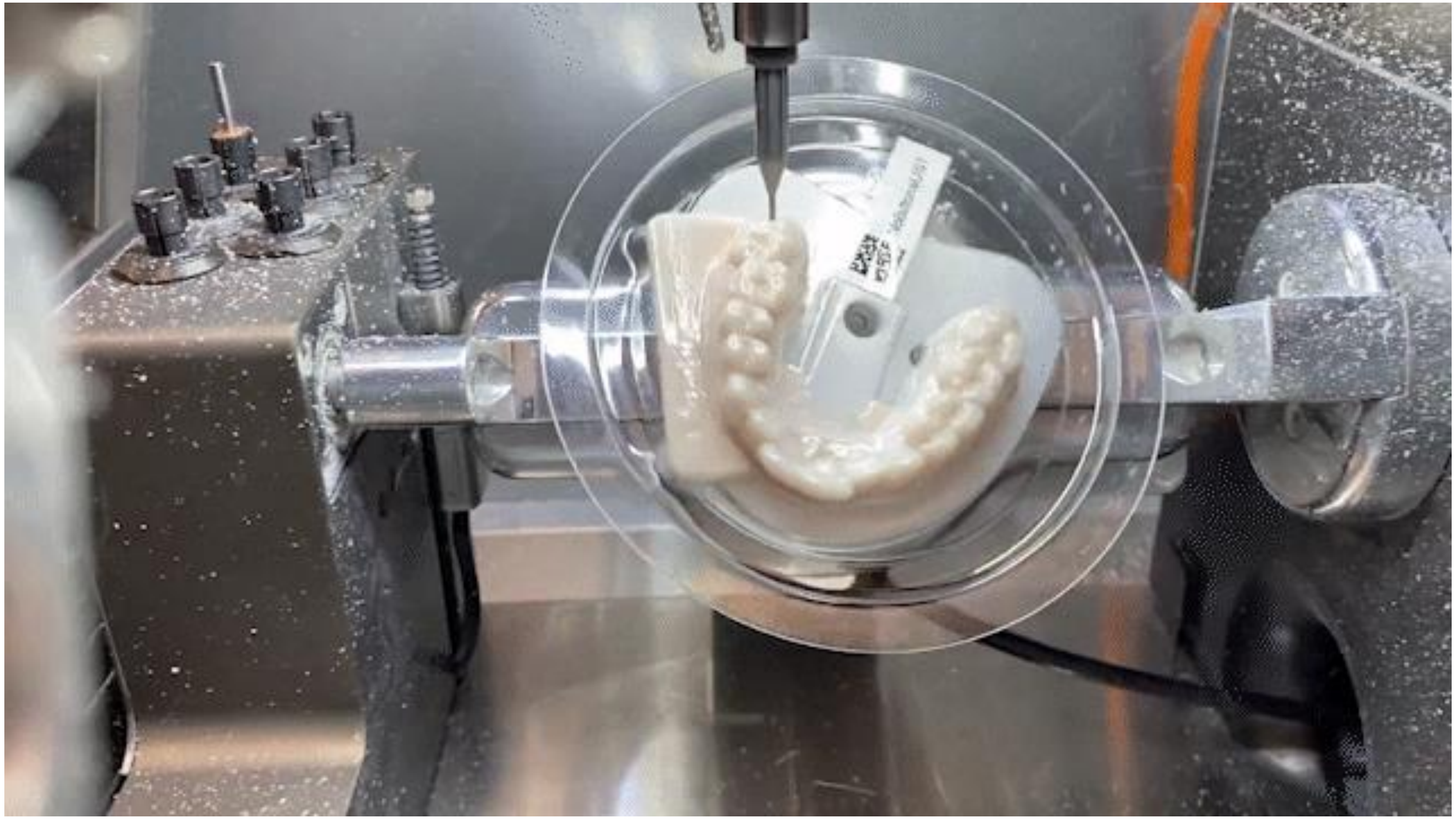
Widely used in metal cutting industry and are best used to produce following types of products,

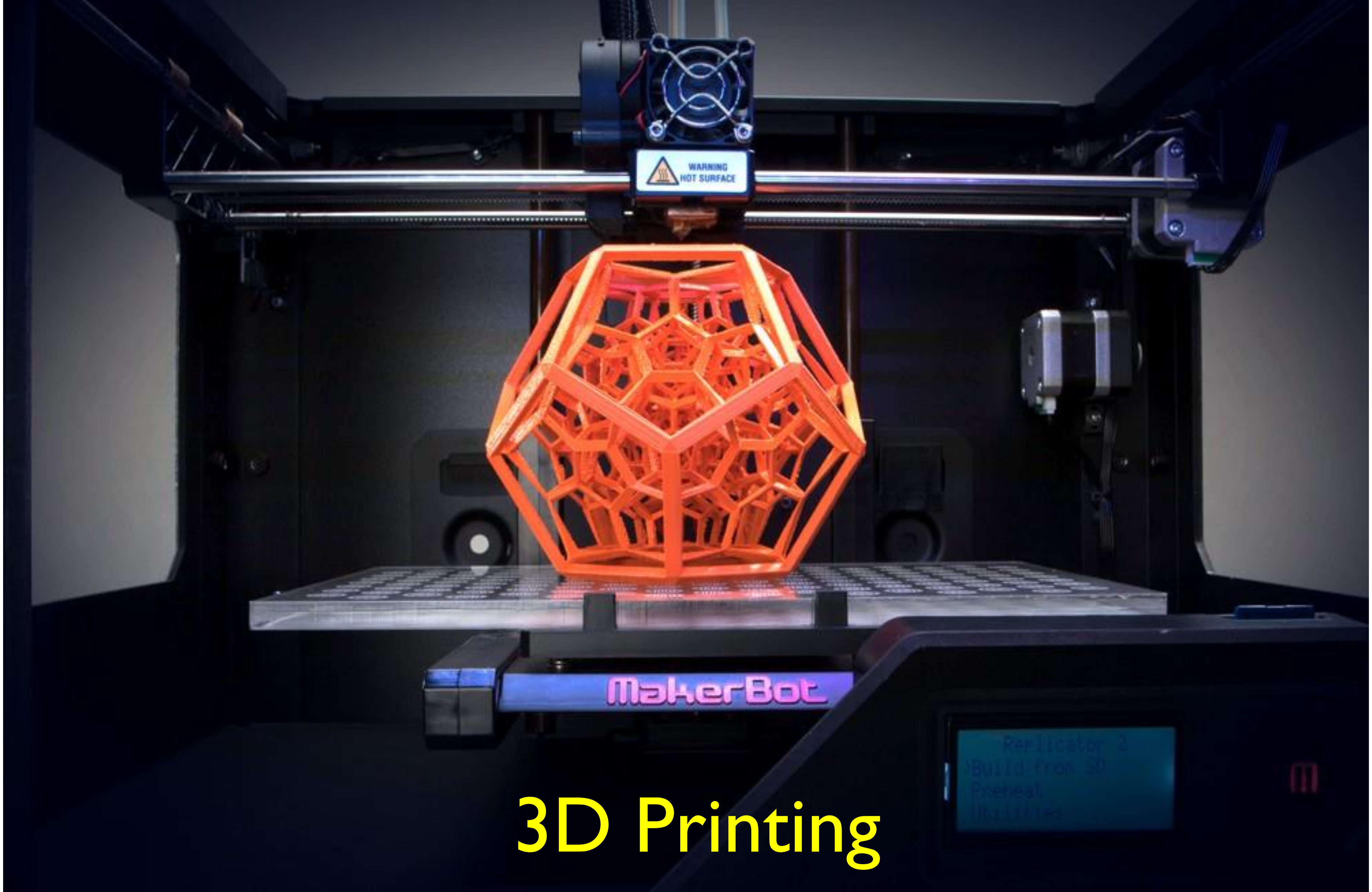
- Parts with complicated contours
- Parts requiring close tolerances and good repeatability
- Incase human errors could be extremely costly
- Small batch lots or short

Also used in medical field

- Making of dental crowns and implants

Application: Medical domain





3D Printing

Special processing and assembly technologies:- Rapid Prototyping

- Rapid Prototyping:- Is a collection of processes used to fabricate a model, part, or tool in minimum possible time

- Rapid prototyping (RP) is a family of fabrication methods to make engineering prototypes in minimum possible lead times based on a computer-aided design (CAD) model of the item

- A number of rapid prototyping techniques are now available that allow a part to be produced in hours or days rather than weeks, given that a computer model of the part has been generated on a CAD system

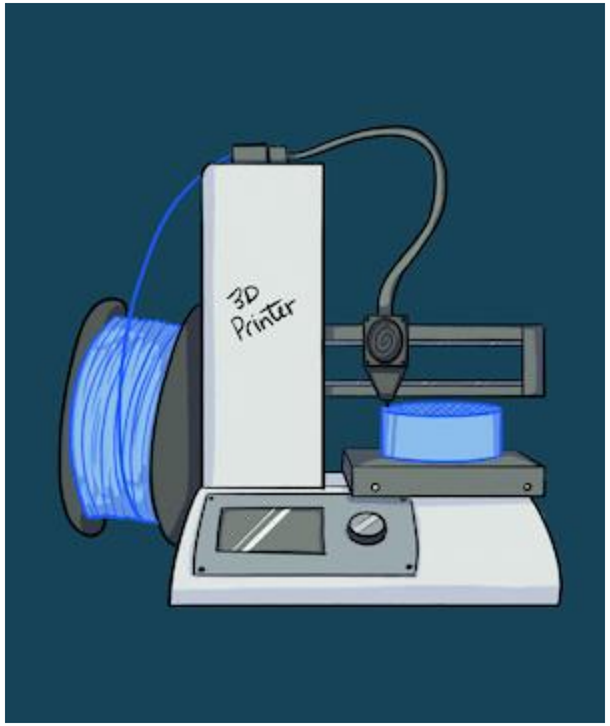
Fundamentals of Rapid Prototyping

- Rapid prototyping technologies arises because product designers would like to have a physical model of a new part or product design rather than a computer model or line drawing
- A virtual prototype, which is a computer model of the part design on a CAD system, may not be adequate for the designer to visualize the part
- It certainly is not sufficient to conduct real physical tests on the part, although it is possible to perform simulated tests by finite element analysis or other methods
- Using one of the available RP technologies, a solid physical part can be created in a relatively short time
- The designer can therefore visually examine and physically feel the part and begin to perform tests and experiments to assess its merits and shortcomings.

Rapid Prototyping

- Two types
  1) Material removal process
  2) Material addition process
- Material removal RP:- machining, milling, drilling with CNC etc.
- The starting material is often a solid block of wax/wood/plastic, which is very easy to machine, and the part and chips can be melted and resolidified for reuse when the current prototype is no longer needed
- Material-addition RP technologies, all of which work by adding layers of material one at a time to build the solid part from bottom to top
- Starting materials include
  a) liquid monomers that are cured layer by layer into solid polymers,
  b) powders that are aggregated and bonded layer by layer, and
  c) solid sheets that are laminated to create the solid part

Material Removal RP



Material Addition RP

3D Printed House!

3D Printed House in IITM

Rapid Prototyping

- Various material addition methods are used
- Some techniques use lasers to solidify the starting material, another deposits a soft plastic filament in the outline of each layer, while others bond solid layers together

Applications

Biomedical applications

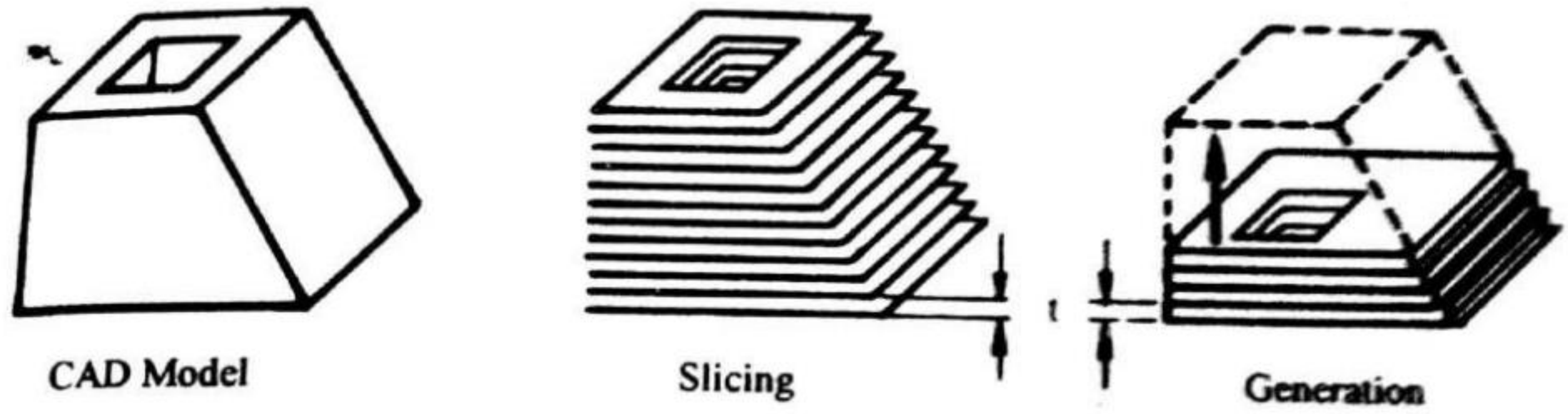
Applications

3D Printed steel bridge in Amsterdam

Limitations of RP

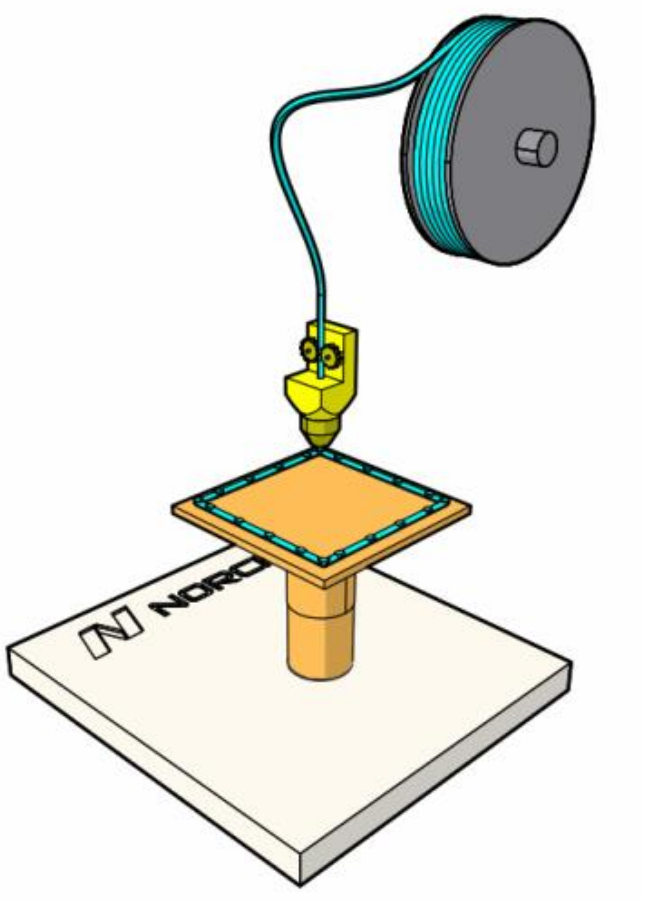
- Limited variety of materials
- Part accuracy
- Mechanical performance of materials
- Error associated with process:- Mathematical, process related, and material related
- In general, the materials used in RP systems are not as strong as the production part materials that will be used in the actual product. This limits the mechanical performance of the prototypes and the amount of realistic testing that can be done to verify the design during product development.

Basic Principle of RP



RP:- Fused Deposition Modeling

- Filament of wax or polymer is extruded onto the surface from a workhead to complete each new layer
- Workhead moves in x-y direction for each layer, and for new year in z-direction
- Workhead heats the material to about 0.5^{o}C above its melting point before extruding it onto the part surface



Thank You